Yavuz Shahzad

Montreal, QC yavuz.shahzad@mail.mcgill.ca +1-438-874-3985 yavuzshahzad.com yavuz-shahzad

Skills

Programming Languages: Python, R, SQL, LaTeX.
Libraries and Frameworks: Polars, Pandas, PySpark, Scikit-learn, Seaborn, Tidyverse.
Developer Tools and Platforms: GitHub, Databricks, Azure, Jupyter Notebook, Excel.

Education

McGill University ***Montreal, QC***
MSc in Biostatistics *2026 – 2028*

- **Research area:** Causal machine learning for heterogeneous treatment effect estimation.

McGill University ***Montreal, QC***
BA in Statistics, Minor in Psychology *2022 – 2026*

- **GPA:** 3.68/4.0
- **Awards:** NSERC Undergraduate Student Research Award (2024, 2026), Dean's Honour List (2022).
- **Coursework:** Mathematical Statistics I & II, Experimental Design, Sampling Theory, Generalized Linear Models, Statistical Learning, Honours Analysis I & II, Honours Algebra I & II.

Experience

Undergraduate Researcher ***Montreal, QC***
McGill Clinical and Health Informatics Group *Aug 2025 – Sep 2026*

- Developed probabilistic regression models (NGBoost) for uncertainty-aware prediction of individualized hemoglobin recovery after blood donation.
- Engineered a novel calibration method for probabilistic regression models, enabling accurate quantification of low hemoglobin deferral risk.

Data Science Intern ***Toronto, ON***
Manulife *May 2025 – Aug 2025*

- Developed time series models (Prophet) to forecast new regular premiums for life insurance products, reducing MAPE by 45% compared to legacy models.
- Conducted exploratory data analysis to recommend new lead types for proactive outreach to financial advisors and implemented the delivery of leads to wholesalers in a production PySpark pipeline.
- Presented leads pipeline and forecasting models to senior leadership, incorporating stakeholder feedback.

Undergraduate Researcher ***Montreal, QC***
Brain Research and Imaging of Neurorehabilitation Lab: McGill University *Jan 2023 – Sep 2024*

- Cleaned and analyzed data, conducted literature review, and drafted a manuscript about a novel treatment paradigm for those with chronic stroke.
- Analyzed neural markers of long-COVID survivors using a threshold-based β-burst detection algorithm.

Research Intern ***Boston, MA***
J. Philip Kistler Stroke Research Center: Harvard Medical School *Jun 2022 – Aug 2022*

- Abstracted electronic health records of 200+ patients to study left atrial appendage closure outcomes.

Publications and Manuscripts

- Milot, M.-H., Palimeris, S., **Shahzad, Y.**, Corriveau, H., Tremblay, F., Boudrias, M.-H.
  Long-term benefits of a tailored strength training intervention on arm function in chronic stroke survivors: a follow-up study. **Journal of Rehabilitation Medicine – Clinical Communications**, Vol. 8, February 2025.
- **Shahzad, Y.**, Li, W., Bravo, M., Custer, B. S., Pham, T. D., Russell, W. A.
  Probabilistic prediction of returning blood donors' hemoglobin levels. In preparation for submission to *Transfusion*.
- An, E., Alavi, N., . . . , **Shahzad, Y.**, . . . , Bartoszko, J.
  Variation in use of thrombin-enhancing factor concentrates in adult cardiac surgery: a multicenter observational analysis. Manuscript in preparation.